Ps 04

Erik An

<2026-01-13 Tue>

Contents

1 Problem 10 (a) DONE 3

1.1 Task 3

1.2 Solution 3

2 Problem 10 (b) DONE 4

2.1 Task 4

2.2 Solution 4

3 Problem 10 (c) DONE 5

3.1 Task 5

3.2 Solution 5

4 Problem 10 (d) DONE 6

4.1 Task 6

4.2 Solution 7

5 Problem 11 (a) DONE 9

5.1 Task 9

5.2 Solution 9

6 Problem 11 (b) DONE 11

6.1 Task 11
6.2 Solution 12

7 Problem 11 (c) DONE 12

7.1 Task 12
7.2 Solution 12

8 Problem 12 (a) DONE 15

8.1 Task 15
8.2 Solution 16

9 Problem 12 (b) DONE 17

9.1 Task 17
9.2 Solution 17

10 Problem 12 (c) DONE 18

10.1 Task 18
10.2 Solution 18

1 Problem 10 (a) DONE

1.1 Task

In this exercise, we work over the field $\mathbb{R}$. Consider $A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 5 & -3 \\ -2 & 5 & -2 \end{bmatrix} \in \mathbb{R}^{3\times 3}$, $b = \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix} \in \mathbb{R}^3$. Perform Gaussian elimination on the system $Ax = b$ without row exchanges. Show explicitly how you eliminate the entries below the diagonal in the first and second columns, and write down the resulting upper-triangular matrix U.

1.2 Solution

$$\begin{bmatrix} 2 & 1 & -1 \\ 4 & 5 & -3 \\ -2 & 5 & -2 \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix}$$

$$[A \mid b] = \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 4 & 5 & -3 & 7 \\ -2 & 5 & -2 & 3 \end{array}\right]$$

- **Step 1: Eliminate Column 1 Pivot**: $a_{11} = 2$.
  - **Row 2 Operation**: $R_2 \leftarrow R_2 - (2) \times R_1$

$$\begin{array}{rl} \text{Row 2:} & [4, 5, -3 \mid 7] \\ -(2) \times \text{Row 1:} & [-4, -2, 2 \mid -2] \\ \hline \text{New Row 2:} & [0, 3, -1 \mid 5] \end{array}$$

  - **Row 3 Operation**: $R_3 \leftarrow R_3 - (-1) \times R_1$

$$\begin{array}{rl} \text{Row 3:} & [-2, 5, -2 \mid 3] \\ -(-1)\text{Row 1:} & [2, 1, -1 \mid 1] \\ \hline \text{New Row 3:} & [0, 6, -3 \mid 4] \end{array}$$

Current Augmented Matrix:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 3 & -1 & 5 \\ 0 & 6 & -3 & 4 \end{array}\right]$$

- **Step 2: Eliminate Column 2 Pivot**: $a_{22} = 3$.
  - **Row 3 Operation**: $R_3 \leftarrow R_3 - (2) \times R_2$

$$\begin{array}{rl} \text{Row 3:} & [0, 6, -3 \mid 4] \\ -(2) \times \text{Row 2:} & [0, -6, 2 \mid -10] \\ \hline \text{New Row 3:} & [0, 0, -1 \mid -6] \end{array}$$

Final Upper Triangular Form:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 3 & -1 & 5 \\ 0 & 0 & -1 & -6 \end{array}\right]$$

Upper Triangular Matrix U:

$$U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

2 Problem 10 (b) DONE

2.1 Task

Let m_{21}, m_{31} be the multipliers used to eliminate the entries in the first column, and let m_{32} be the multiplier used to eliminate the (3, 2)-entry after the first elimination step. Form the unit lower-triangular matrix $L = \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix}$ Verify that your matrices satisfy $A = LU$.

2.2 Solution

From the row operations in Part (a), we identify the multipliers:

$$m_{21} = 2, \quad m_{31} = -1, \quad m_{32} = 2$$

Constructing L:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$$

We verify the decomposition by computing the product LU:

$$
\begin{aligned}
LU &= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -1 \\ 0 & 0 & -1 \end{bmatrix} \\
&= \begin{bmatrix} (1)(2) & (1)(1) & (1)(-1) \\ (2)(2) & (2)(1)+(1)(3) & (2)(-1)+(1)(-1) \\ (-1)(2) & (-1)(1)+(2)(3) & (-1)(-1)+(2)(-1)+(1)(-1) \end{bmatrix} \\
&= \begin{bmatrix} 2 & 1 & -1 \\ 4 & 5 & -3 \\ -2 & 5 & -2 \end{bmatrix} \\
&= A \quad \checkmark
\end{aligned}
$$

3 Problem 10 (c) DONE

3.1 Task

Solve $Ax = b$ using the factorization $A = LU$ by (i) solving $Ly = b$ for y (forward substitution) and then (ii) solving $Ux = y$ for x (backward substitution). Present your intermediate vector y and the final solution x.

3.2 Solution

We solve $Ax = b$ by using the decomposition $A = LU$. This allows us to solve two triangular systems:

$$
\begin{aligned}
Ly &= b \quad (\text{where } y = Ux) \\
Ux &= y
\end{aligned}
$$

- **(i) Forward Substitution** ($Ly = b$)

  We set up the system:

$$
\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix}
$$

  From the first row:

$$
y_1 = 1
$$

  From the second row:

$$
2y_1 + y_2 = 7 \implies 2(1) + y_2 = 7 \implies y_2 = 5
$$

From the third row:

$$-1y_1 + 2y_2 + y_3 = 3 \implies -1(1) + 2(5) + y_3 = 3 \implies 9 + y_3 = 3 \implies y_3 = -6$$

The intermediate vector is:

$$y = \begin{bmatrix} 1 \\ 5 \\ -6 \end{bmatrix}$$

- **(ii) Backward Substitution** ($Ux = y$) We now solve for x:

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -6 \end{bmatrix}$$

From the third row:

$$-1x_3 = -6 \implies x_3 = 6$$

From the second row:

$$3x_2 - x_3 = 5 \implies 3x_2 - 6 = 5 \implies 3x_2 = 11 \implies x_2 = \frac{11}{3}$$

From the first row:

$$2x_1 + x_2 - x_3 = 1 \implies 2x_1 + \frac{11}{3} - 6 = 1$$

$$2x_1 - \frac{7}{3} = 1 \implies 2x_1 = \frac{10}{3} \implies x_1 = \frac{5}{3}$$

Final Solution

$$x = \begin{bmatrix} 5/3 \\ 11/3 \\ 6 \end{bmatrix}$$

4 Problem 10 (d) DONE

4.1 Task

(Programming, Julia) Implement two Julia functions `forward_substitution(L,b)` and `backward_substitution(U,y)` that solve $Ly = b$ band $Ux = y$ for triangular matrices of compatible sizes. Use your code to solve the system from this exercise and verify that your result agrees with Julia's A b.

4.2 Solution

```
is_compatible(m_1, m_2) = size(m_1, 2) == size(m_2, 1)

multiply(row, column) = row' * column

function forward_substitution(L, b)
    if !is_compatible(L, b)
        error("Matrix Dimensions Mismatch.")
    end

    n = size(L, 1)

    x = zeros(Float64, n)

    for i in 1:n
        sum_val = b[i]
        for j in 1:(i-1)
            sum_val -= L[i, j] * x[j]
        end
        x[i] = sum_val / L[i, i]
    end
    return x
end

function backward_substitution(U, y)
    if !is_compatible(U, y)
        error("Matrix Dimensions Mismatch.")
    end

    n = size(U, 1)

    x = zeros(Float64, n)

    for i in n:-1:1
        sum_val = y[i]
        for j in (i+1):n
            sum_val -= U[i, j] * x[j]
        end
        x[i] = sum_val / U[i, i]
    end
    return x
end

L = [ 1  0  0;
      2  1  0;
     -1  2  1]
```

```
U = [ 2  1 -1;
      0  3 -1;
      0  0 -1]

b = [ 1;
      7;
      3]

y = forward_substitution(L, b)
x = backward_substitution(U, y)

println("Intermediate vector y: ", y)
println("Final solution x: ", x)
```

```
is_compatible (generic function with 1 method)
multiply (generic function with 1 method)
forward_substitution (generic function with 1 method)
backward_substitution (generic function with 1 method)
3×3 Matrix{Int64}:
  1  0  0
  2  1  0
 -1  2  1
3×3 Matrix{Int64}:
 2  1  -1
 0  3  -1
 0  0  -1
3-element Vector{Int64}:
 1
 7
 3
3-element Vector{Float64}:
  1.0
  5.0
 -6.0
3-element Vector{Float64}:
 1.6666666666666667
 3.6666666666666665
 6.0
Intermediate vector y: [1.0, 5.0, -6.0]
Final solution x: [1.6666666666666667, 3.6666666666666665, 6.0]
```

5 Problem 11 (a) DONE

5.1 Task

Tall system. Consider $U = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{4\times 3}$ $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \in \mathbb{R}^4$ Write $U = \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}$ with $U_1 \in \mathbb{R}^{3\times 3}$ and $U_2 \in \mathbb{R}^{1\times 3}$. Derive a necessary and sufficient condition on y for the system

$$Uz = y$$

to have a solution $z \in \mathbb{R}^3$, and express the (necessarily unique) solution z in terms of y_1, y_2, y_3 Then check your condition for the concrete right-hand side

$$y = \begin{bmatrix} 2 \\ 1 \\ 3 \\ -2 \end{bmatrix}$$

5.2 Solution

- **Step 1: Foundation.**

  Given:

$$U = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{4\times 3} \qquad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \in \mathbb{R}^4$$

$$\begin{aligned} U &= \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} \\ \implies U_1 &= \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix} \\ U_2 &= \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \end{aligned}$$

  Since U_1 is an Upper Triangle Matrix and has non-zero entries on diagonal, it is invertable. Which implies $\exists z : U_1 z = [y_1, y_2, y_3]^\top$ has a unique solution.

- **Step 2: Find Unique z Candidate (Back Substitution).**

  Given:

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

- **Sub-Step 1: Solve for the 3rd Row.**

  $$2z_3 = y_3 \implies z_3 = \frac{1}{2}y_3$$

- **Sub-Step 2: Solve for the 2nd Row.**

  $$\begin{aligned} z_2 + 3z_3 = y_2 &\implies z_2 + 3\left(\frac{1}{2}y_3\right) = y_2 \\ &\implies z_2 + \frac{3}{2}y_3 = y_2 \\ &\implies z_2 = y_2 - \frac{3}{2}y_3 \end{aligned}$$

- **Sub-Step 3: Solve for the 1st Row.**

  $$\begin{aligned} z_1 + 2z_2 - z_3 = y_1 &\implies z_1 + 2\left(y_2 - \frac{3}{2}y_3\right) - \left(\frac{1}{2}y_3\right) = y_1 \\ &\implies z_1 + 2y_2 - 3y_3 - \frac{1}{2}y_3 = y_1 \\ &\implies z_1 + 2y_2 - \frac{7}{2}y_3 = y_1 \\ &\implies z_1 = y_1 - 2y_2 + \frac{7}{2}y_3 \end{aligned}$$

Hence, the final candidate solution:

$$z = \begin{bmatrix} y_1 - 2y_2 + \frac{7}{2}y_3 \\ y_2 - \frac{3}{2}y_3 \\ \frac{1}{2}y_3 \end{bmatrix}$$

- **Step 3: Derive the Condition ($U_2 z = y_4$).**

  For a system to have a solution, the candidate z must satisfy:

  $$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = y_4$$

  Which simplifies to:

  $$z_2 + z_3 = y_4$$

  $$y_2 - \frac{3}{2}y_3 + \frac{1}{2}y_3 = y_4 \quad (z_2, z_3 \text{ substitution})$$

  $$\implies y_2 - y_3 = y_4$$

The Necessary and Sufficient Condition:

$$y_2 - y_3 - y_4 = 0$$

- **Step 4: Validate with the Vector y.**
  Given:

$$y = \begin{bmatrix} 2 \\ 1 \\ 3 \\ -2 \end{bmatrix}$$

  Indexing:

$$\begin{aligned} y_1 &= 2 \\ y_2 &= 1 \\ y_3 &= 3 \\ y_4 &= -2 \end{aligned}$$

  - **Validate the Condition**:

$$y_2 - y_3 - y_4 = 1 - 3 - (-2) = 0$$

    The condition holds.
  - **Calculate Solution**:

$$\begin{aligned} z_3 &= \frac{1}{2}y_3 = \frac{1}{2} \cdot 3 = \frac{3}{2} \\ z_2 &= y_2 - \frac{3}{2}y_3 = 1 - \frac{3}{2} \cdot 3 = 1 - \frac{9}{2} = -\frac{7}{2} \\ z_1 &= y_1 - 2y_2 + \frac{7}{2}y_3 = 2 - 2 \cdot 1 + \frac{7}{2} \cdot 3 = \frac{21}{2} \end{aligned}$$

  - **Final Solution**:

$$z = \begin{bmatrix} \frac{21}{2} \\ -\frac{7}{2} \\ \frac{3}{2} \end{bmatrix}$$

6 Problem 11 (b) DONE

6.1 Task

Discuss why the solution to the system $Uz = y$ is necessarily unique.

6.2 Solution

- 1. **Uniqueness of a Sub-system.**

  Since the sub-matrix U_1 is upper-triangular with non-zero entries on diagonal, the columns of U_1 are linearly independent. Thus, the system $U_1 z = [y_1, y_2, y_3]^\top$ holds for exactly one unique solution, which I will further call z^*.

- 2. **The Constraint Argument.**

  The solution set for a whole system $Uz = y$ must be a subset of the solution set for the first three rows.

  - The first three rows force the solution z to be exaclty z^*.
  - The fourth row (U_4) acts as a contraint.
  - Thus, the whole system has either **exactly one** solution (if z^* satisfies the fourth row) or **no** solution (if it does not).
  - It is impossible for it to have infinite solutions or a second solution, because any solution must be z^*.

Summary: Because the columns of the top block U_1 are linearly independent, the columns of the entire matrix U are also linearly independent. A matrix with linearly independent columns always yields a necessarily unique solution (if one exists).

7 Problem 11 (c) DONE

7.1 Task

Wide system. Consider $W = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & -1 & 3 \end{bmatrix} \in \mathbb{R}^{2\times 4}$; $z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \in \mathbb{R}^2$. Partition $W = \begin{bmatrix} W_1 & W_2 \end{bmatrix}$ with $W_1 \in \mathbb{R}^{2\times 2}$ and $W_2 \in \mathbb{R}^{2\times 2}$ and similarly split $x = \begin{bmatrix} x^{(1)} \\ x^{(2)} \end{bmatrix}$ with $x^{(1)} \in \mathbb{R}^2$ and $x^{(2)} \in \mathbb{R}^2$. Express the full solution set $\{x \in \mathbb{R}^4 : Wx = z\}$ in terms of the free variables $x^{(2)}$. For the concrete right-hand side $z = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ provide two distinct solutions.

7.2 Solution

- **Step 1: Foundation.**

  Given:

$$
\begin{aligned}
W &= \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & -1 & 3 \end{bmatrix}, \quad z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \\
W &= [W_1, W_2] \\
\implies W_1 &= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \\
W_2 &= \begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
x &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \\
x &= \begin{bmatrix} x^{(1)} \\ x^{(2)} \end{bmatrix} \\
\implies x^{(1)} &= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
x^{(2)} &= \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}
\end{aligned}
$$

Equation given:

$$
\begin{aligned}
Wx &= z \\
\begin{bmatrix} W_1 & W_2 \end{bmatrix} \begin{bmatrix} x^{(1)} \\ x^{(2)} \end{bmatrix} &= z \quad \text{(substituting partitions)} \\
W_1 x^{(1)} + W_2 x^{(2)} &= z \\
W_1 x^{(1)} &= z - W_2 x^{(2)} \quad (\text{as } z \in \mathbb{R}^2 \wedge W_2 x^{(2)} \in \mathbb{R}^2)
\end{aligned}
$$

By the problem definition, I consider $x^{(2)}$ as a *free variables*. Thus, let $x^{(2)} = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}$.

Since W_1 is an Upper Triangular Matrix and has non-zero entries on diagonal, it is invertable. It implies that for any fixed $x^{(2)}$, there exists a unique corresponding $x^{(1)}$. The "uniqueness" is relative to the choice of the free variables.

- **Step 2: Derivation of the General Solution in Terms of Free Variables.**

$$
\begin{aligned}
W_1 x^{(1)} &= z - W_2 x^{(2)} \\
&= \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} \\
&= \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} - \begin{bmatrix} x_3 \\ -x_3 + 3x_4 \end{bmatrix} \\
&= \begin{bmatrix} z_1 - x_3 \\ z_2 + x_3 - 3x_4 \end{bmatrix}
\end{aligned}
$$

Substituting W_1 and $x^{(1)}$:

$$
\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} z_1 - x_3 \\ z_2 + x_3 - 3x_4 \end{bmatrix}
$$

- **Sub-Step 1: Solve for the 2nd Row.**

$$
x_2 = z_2 + x_3 - 3x_4
$$

- **Sub-Step 2: Solve for the 1st Row.**

$$
\begin{aligned}
x_1 + 2x_2 = z_1 - x_3 &\implies x_1 + 2(z_2 + x_3 - 3x_4) = z_1 - x_3 \\
&\implies x_1 = z_1 - x_3 - 2(z_2 + x_3 - 3x_4) \\
&\implies x_1 = z_1 - x_3 - 2z_2 - 2x_3 + 6x_4 \\
&\implies x_1 = z_1 - 2z_2 - 3x_3 + 6x_4
\end{aligned}
$$

Hence, the final candidate solution:

$$
x^{(1)} = \begin{bmatrix} z_1 - 2z_2 - 3x_3 + 6x_4 \\ z_2 + x_3 - 3x_4 \end{bmatrix}
$$

- **Step 3: Calculation of Distinct Particular Solutions.**

  Full solution vector:

$$
x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} z_1 - 2z_2 - 3x_3 + 6x_4 \\ z_2 + x_3 - 3x_4 \\ x_3 \\ x_4 \end{bmatrix}
$$

  Given:

$$
\begin{aligned}
& z = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \\
\implies & z_1 = 5 \\
& z_2 = 1
\end{aligned}
$$

- Case $x_A^{(2)} = [6, 7]^\top$ (arbitrary chosen):

$$\begin{aligned} x^{(1)} &= \begin{bmatrix} z_1 - 2z_2 - 3x_3 + 6x_4 \\ z_2 + x_3 - 3x_4 \end{bmatrix} \\ &= \begin{bmatrix} 5 - 2(1) - 3(6) + 6(7) \\ 1 + 6 - 3(7) \end{bmatrix} \\ &= \begin{bmatrix} 5 - 2 - 18 + 42 \\ 1 + 6 - 21 \end{bmatrix} \\ &= \begin{bmatrix} 27 \\ -14 \end{bmatrix} \end{aligned}$$

- Case $x_B^{(2)} = [6, 9]^\top$ (arbitrary chosen):

$$\begin{aligned} x^{(1)} &= \begin{bmatrix} z_1 - 2z_2 - 3x_3 + 6x_4 \\ z_2 + x_3 - 3x_4 \end{bmatrix} \\ &= \begin{bmatrix} 5 - 2(1) - 3(6) + 6(9) \\ 1 + 6 - 3(9) \end{bmatrix} \\ &= \begin{bmatrix} 5 - 2 - 18 + 54 \\ 1 + 6 - 27 \end{bmatrix} \\ &= \begin{bmatrix} 39 \\ -20 \end{bmatrix} \end{aligned}$$

Hence, full distinct solutions for both cases:

$$x_A = \begin{bmatrix} 27 \\ -14 \\ 6 \\ 7 \end{bmatrix}, \quad x_B = \begin{bmatrix} 39 \\ -20 \\ 6 \\ 9 \end{bmatrix}$$

8 Problem 12 (a) DONE

8.1 Task

We work over $\mathbb{R}$. Consider $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 4 \end{bmatrix}$ $b = \begin{bmatrix} 3 \\ 1 \\ 8 \end{bmatrix}$ Let Π be the permutation matrix that exchanges the first and second rows. Write Π explicitly and verify (by a direct calculation) that $\Pi^{-1} = \Pi^T$.

8.2 Solution

- **Step 1: Foundation and Dimensions.**

  The permutation matrix Π operates on $A \in \mathbb{R}^{3\times 3}$ and $b \in \mathbb{R}^3$. Therefore, to satisfy matrix multiplication dimensions, Π must be in $\mathbb{R}^{3\times 3}$.

  We construct Π by taking the Identity matrix I_3 and swapping the first and second rows:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
$$\implies \Pi = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

  We obtain the transpose $\Pi^\top$ by swapping rows and columns (which leaves Π unchanged in this specific case):

$$\Pi^\top = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- **Step 2: Verification by Direct Calculation.**

  To verify that $\Pi^{-1} = \Pi^\top$, we must show that $\Pi\Pi^\top = I$.

$$\Pi\Pi^\top = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

  Calculating the diagonal entries as dot products of rows and columns:

  - Row 1 $\cdot$ Col 1: $(0)(0) + (1)(1) + (0)(0) = 1$
  - Row 2 $\cdot$ Col 2: $(1)(1) + (0)(0) + (0)(0) = 1$
  - Row 3 $\cdot$ Col 3: $(0)(0) + (0)(0) + (1)(1) = 1$

  The off-diagonal entries are clearly 0 (orthogonal vectors). Thus:

$$\Pi\Pi^\top = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

- **Step 3: Conclusion.**

  Since $\Pi\Pi^\top = I$, by the definition of the matrix inverse, it follows that $\Pi^{-1} = \Pi^\top$.

9 Problem 12 (b) DONE

9.1 Task

Prove that the linear systems $Ax = b$ and $(\Pi A)x = \Pi b$ have exactly the same solution set.

9.2 Solution

To prove the solution sets are identical, we must show that $Ax = b \iff (\Pi A)x = \Pi b$. We do this by proving implication in both directions.

- **Part 1: Forward Direction ($\Longrightarrow$)**

  Assume x satisfies $Ax = b$.

  Multiplying both sides by Π on the left:

$$\begin{aligned} \Pi(Ax) &= \Pi b \\ (\Pi A)x &= \Pi b \quad \text{(multiplicative associativity)} \end{aligned}$$

  Thus, any solution x to the first system is also a solution to the second system.

- **Part 2: Reverse Direction ($\Longleftarrow$)**

  Assume x satisfies $(\Pi A)x = \Pi b$.

  Since Π is a permutation matrix, it is invertible (as verified previously, $\Pi^{-1} = \Pi^T$ exists). We multiply both sides by Π^{-1} on the left:

$$\begin{aligned} \Pi^{-1}(\Pi A)x &= \Pi^{-1}\Pi b \\ (IA)x &= Ib \quad \text{(associativity and identity property } (\Pi^{-1}\Pi = I)) \\ Ax &= b \end{aligned}$$

  Thus, any solution x to the second system is necessarily a solution to the first system.

- **Conclusion**

  Since solutions to the first system imply the second, and solutions to the second imply the first, the solution sets are identical.

10 Problem 12 (c) DONE

10.1 Task

Compute an LU factorization of ΠA without further row exchanges, and use it to solve $Ax = b$ (by solving $(\Pi A)x = \Pi b$ via forward/back substitution).

10.2 Solution

- **Step 1: Apply Permutation Π.**

  We apply the permutation Π (swapping Row 1 and Row 2) to both A and b:

$$\Pi A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

$$\Pi b = \begin{bmatrix} 1 \\ 3 \\ 8 \end{bmatrix}$$

- **Step 2: LU Factorization of ΠA.**

  We reduce ΠA to Upper Triangular form (U) to find the multipliers for L.

  1. **Column 1 Elimination**:
     - Pivot is at $(1, 1)$ with value 1.
     - Entry at $(2, 1)$ is 0. Multiplier $l_{21} = 0$.
     - Entry at $(3, 1)$ is 2. Multiplier $l_{31} = \frac{2}{1} = 2$.
     - Operation: $R_3 \leftarrow R_3 - 2R_1$.

$$R_3 = [2, 3, 4] - [2, 2, 0] = [0, 1, 4]$$

     Current State:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 4 \end{bmatrix}$$

  2. **Column 2 Elimination**:
     - Pivot is at $(2, 2)$ with value 2.
     - Entry at $(3, 2)$ is 1. Multiplier $l_{32} = \frac{1}{2}$.
     - Operation: $R_3 \leftarrow R_3 - \frac{1}{2}R_2$.

$$R_3 = [0, 1, 4] - [0, 1, \frac{1}{2}] = [0, 0, \frac{7}{2}]$$

3. **Resulting Matrices**:

$$U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{7}{2} \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & \frac{1}{2} & 1 \end{bmatrix}$$

- **Step 3: Solve for c (Forward Substitution: $Lc = \Pi b$).**

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 8 \end{bmatrix}$$

$$\begin{aligned} c_1 &= 1 \\ c_2 &= 3 \\ 2c_1 + \frac{1}{2}c_2 + c_3 &= 8 \\ 2(1) + \frac{3}{2} + c_3 &= 8 \\ \frac{7}{2} + c_3 &= \frac{16}{2} \implies c_3 = \frac{9}{2} \end{aligned}$$

$$c = \begin{bmatrix} 1 \\ 3 \\ \frac{9}{2} \end{bmatrix}$$

- **Step 4: Solve for x (Back Substitution: $Ux = c$).**

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{7}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ \frac{9}{2} \end{bmatrix}$$

  - **Row 3**:

$$\frac{7}{2}x_3 = \frac{9}{2} \implies x_3 = \frac{9}{7}$$

  - **Row 2**:

$$2x_2 + x_3 = 3 \implies 2x_2 + \frac{9}{7} = 3$$

$$2x_2 = \frac{21}{7} - \frac{9}{7} = \frac{12}{7} \implies x_2 = \frac{6}{7}$$

  - **Row 1**:

$$x_1 + x_2 = 1 \implies x_1 + \frac{6}{7} = 1 \implies x_1 = \frac{1}{7}$$

- **Final Solution**

$$x = \begin{bmatrix} \frac{1}{7} \\ \frac{6}{7} \\ \frac{9}{7} \end{bmatrix}$$